Single Linked List – Memory Allocation

We have seen:

```
Node *newNode = (Node *)malloc(sizeof(Node));
```

→ *The size of newNode will be equal to the size of a Node structure and newNode is declared as a pointer to Node structure.*

→ *The size of a pointer (newNode) is typically 8 bytes on most modern 64 – bit systems, regardless of the size of the structure it points to.*

→ *The size of the Node structure itself is determined by the sum of the sizes of its members*:

→ *data(an integer) is typically 4 bytes.*

→ *next(a pointer to another Node) is typically 8 bytes.*

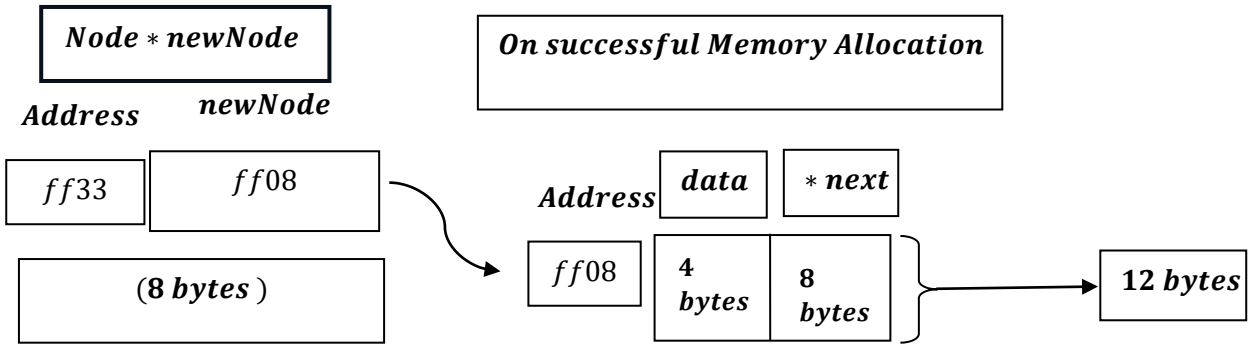
→ *Therefore, the size of the Node structure is typically 12 bytes(4 bytes for data + 8 bytes for next).*

```
Node *newNode = (Node *)malloc(sizeof(Node));
```

What does it mean?

→ *newNode will try to allocate memory for data i.e. 4 bytes, next pointer variable i.e. typically 8 bytes, i.e. total 12 bytes, as discussed earlier, the size of a pointer (newNode) is typically 8 bytes on most modern 64 – bit systems, regardless of the size of the structure it points to.*

Say,



The above is Theoretical , But if we do a reality check:

Theoretical vs. Real-World Allocation

In a real programming environment $\begin{pmatrix} \textbf{like the C++ environments} \\ \textbf{you typically work in} \end{pmatrix}$

, the allocation might not be exactly 12 bytes because of

how CPUs access memory:

- ***Memory Alignment***: *Most* $64-$ *bit compilers align data to* $8-$ *byte boundaries to improve performance.*

- ***Padding***: *To align the* $8-$ *byte* $_{next}$ *pointer, the compiler will often*

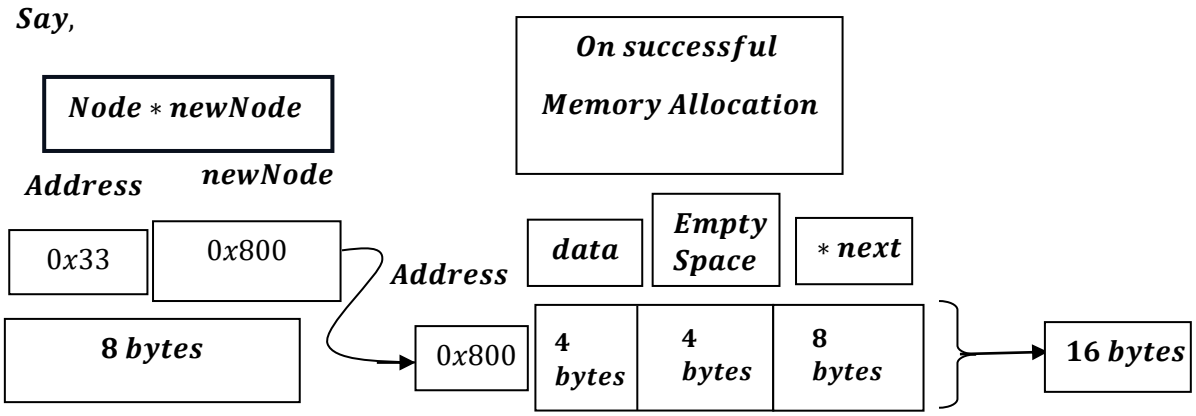
add 4 *bytes of "padding" after the* $4-$ *byte* $_{data}$ *integer.*

- ***Actual Size***: *Consequently,* $_{sizeof}(_{Node})$ *would likely return*

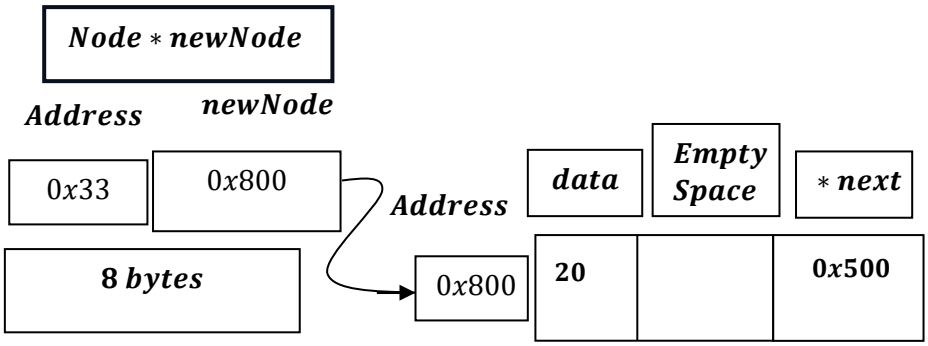
  16 *bytes on your system, rather than the* 12 *bytes.*

Offset	*Size*	*Content*	*Description*
0	**4 *bytes***	***data***	***The integer value*** $(e.g., \mathbf{2})$.
4	**4 *bytes***	***Padding***	***Empty space added by the compiler.***
8	**8 *bytes***	***next***	***The next address of the Node in the linked list*** $(e.g., \mathbf{0x500})$.

It will look like:



Like:



$In\ a\ 64 - bit\ environment, the\ compiler\ aligns\ memory\ to\ the\ size\ of$
$the\ largest\ primitive\ member\ in\ a\ structure - in\ this\ case, the$
$8 - byte\ next\ pointer.$

$Because\ the\ data\ integer\ only\ occupies\ 4\ bytes, the\ compiler\ inserts$
empty$space\ (padding) to\ ensure\ the\ pointer\ starts\ at\ an\ address$
$that\ is\ a\ multiple\ of\ 8.$

$Such\ as:$
$0x500 = 5 \times 16^2 + 0 \times 16^1 + 0 \times 16^0 = 5 \times 256 = 1280$
$1280 \div 8 = 160\ (no\ remainder)$